Currency Analytics Platform: Executive Report

A Cloud-Native Solution for Automated FX Intelligence

1.0 Executive Summary

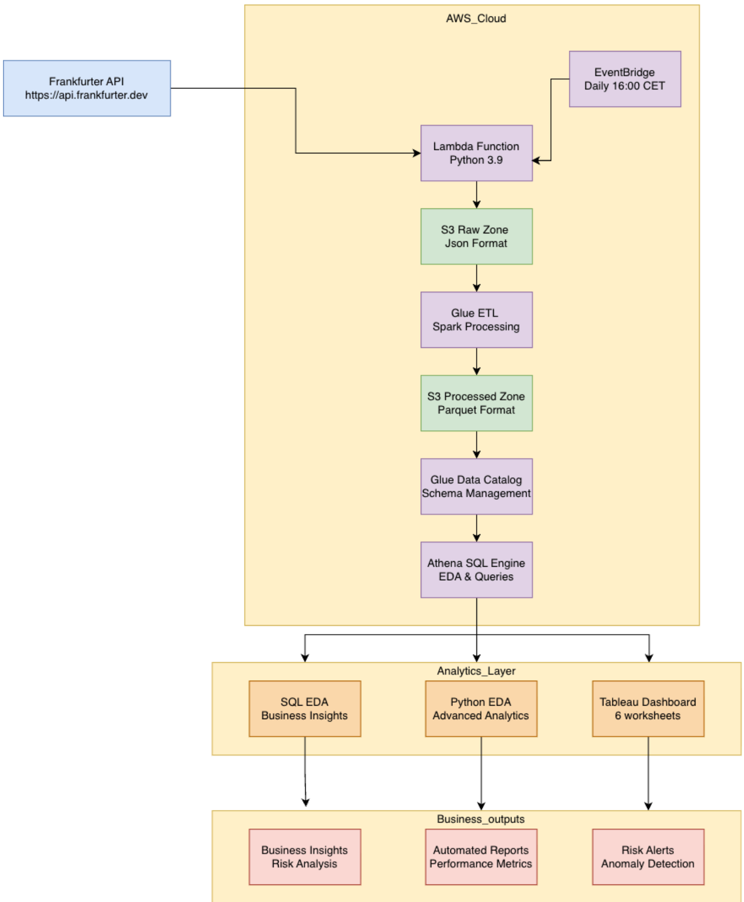
The Currency Analytics Platform is a fully automated, cloud-native system that transforms raw foreign exchange data from the Frankfurter API (https://frankfurter.dev/) into actionable trading insights and risk intelligence.

Manual currency analysis is time-consuming, inefficient, and prone to error. This platform solves that by providing traders, analysts, and portfolio managers with real-time, data-driven analytics.

2.0 Architecture Overview

The platform is built on a fully automated, serverless AWS pipeline, ensuring scalability, cost-efficiency, and minimal operational overhead.

2.1 System Architecture Diagram
(A visual representation of the complete system flow, from data ingestion to visualization.)



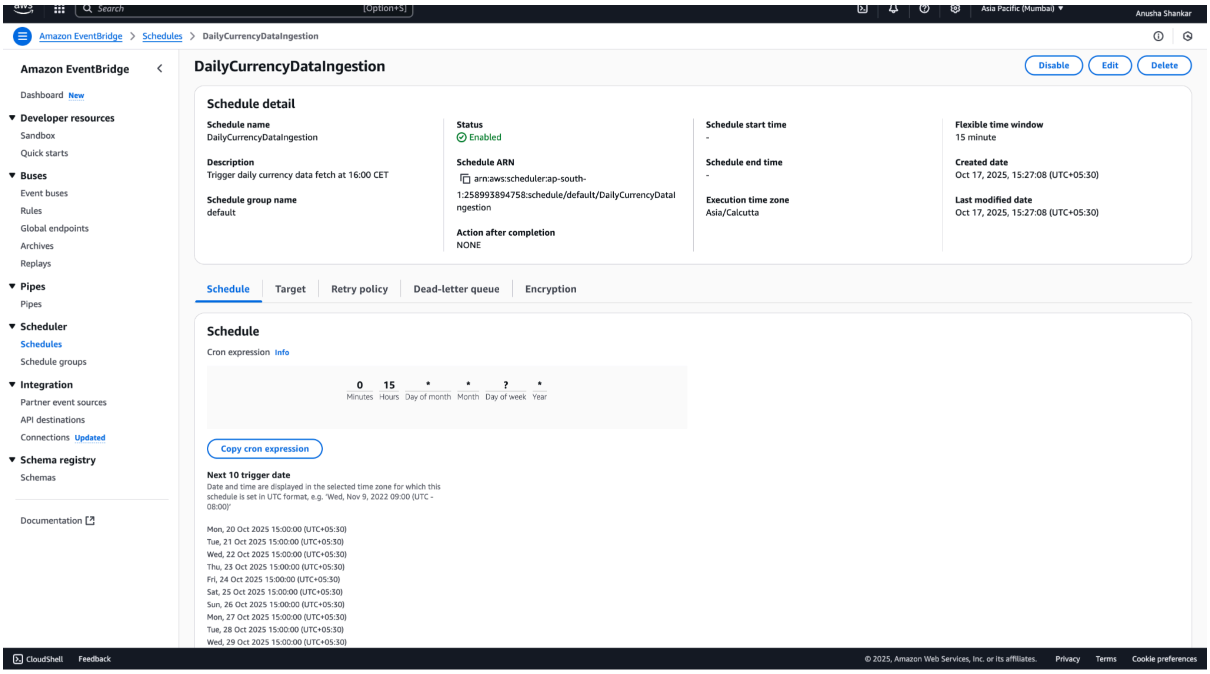
2.2 Technical Implementation & Security

Layer	Technology	Purpose
Ingestion	AWS Lambda (Python)	Daily API data collection
Storage	Amazon S3	Data Lake (Raw & Processed)
Processing	AWS Glue (Spark)	ETL & Metric Calculation
Catalog	AWS Glue Catalog	Schema Management
Crawlers	AWS Glue Crawlers	Auto-discover Raw Data Schema
Query	AWS Athena	SQL Analytics Engine
Scheduling	EventBridge	Automated Daily Pipeline
Visualization	Tableau	Interactive Dashboards

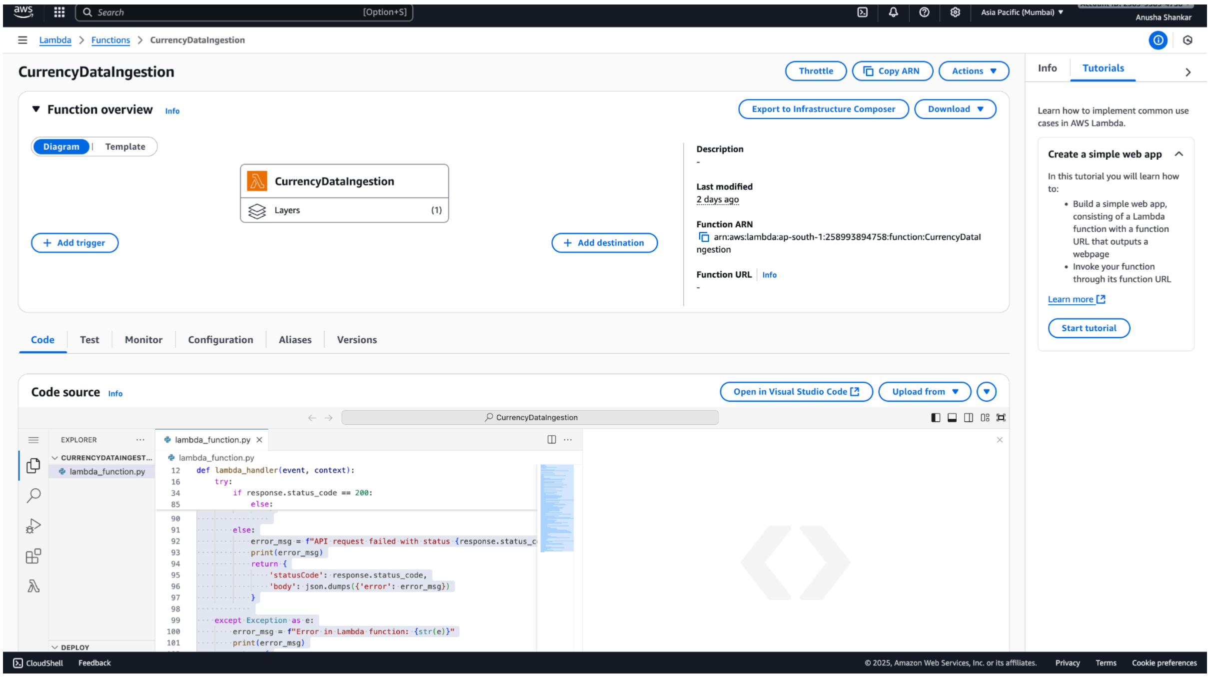
2.3 Data Pipeline Steps

Phase 1: Data Ingestion (Daily)

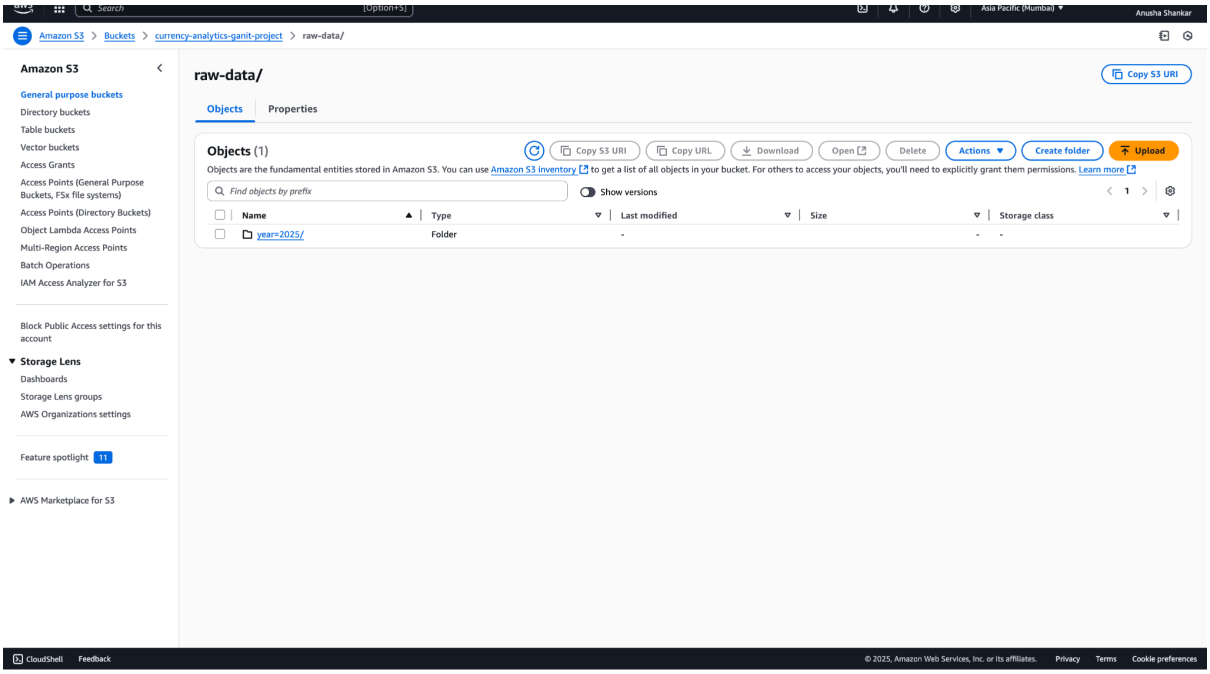
1) Trigger: EventBridge daily at 16:00 CET.



2) Process: Lambda function fetches historical rates from Frankfurter API.

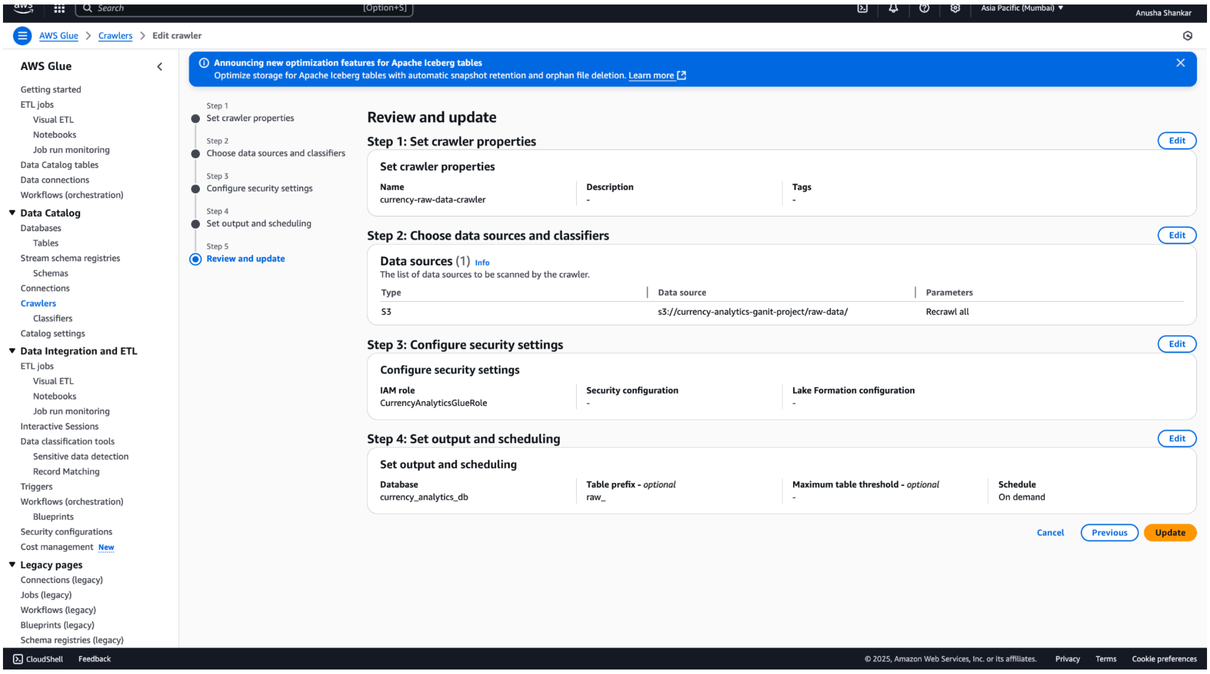


3) Output: JSON files stored in S3, partitioned by year/month.

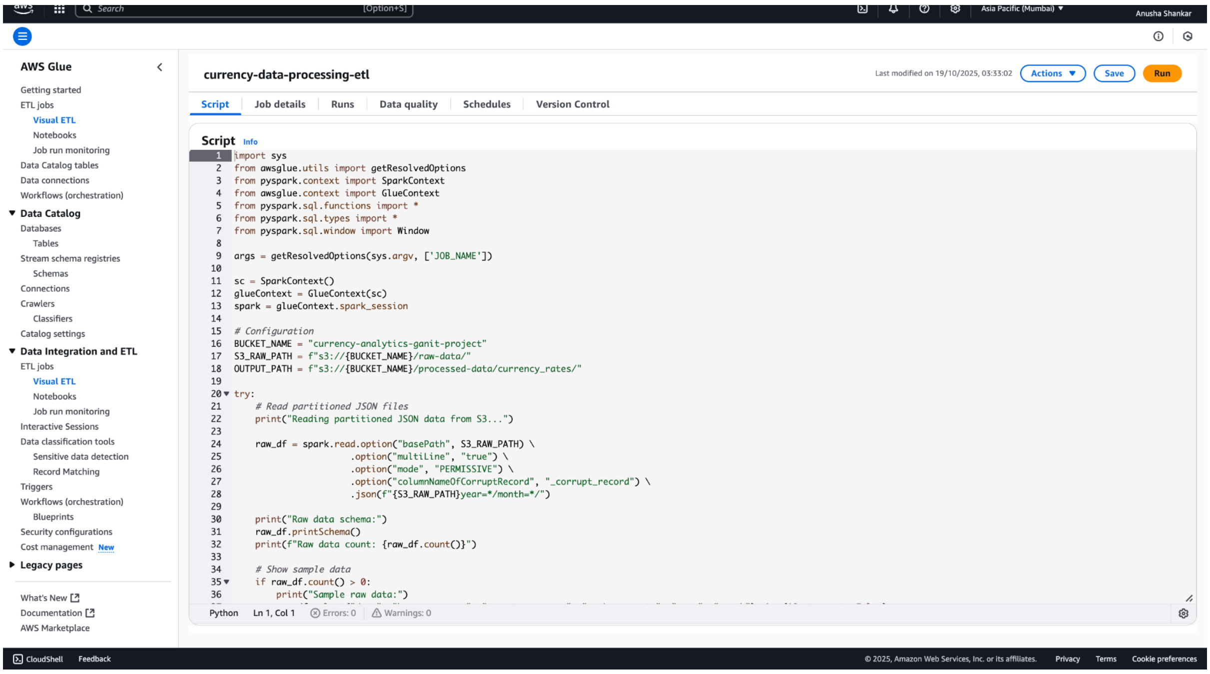


Phase 2: Data Processing (Automated)

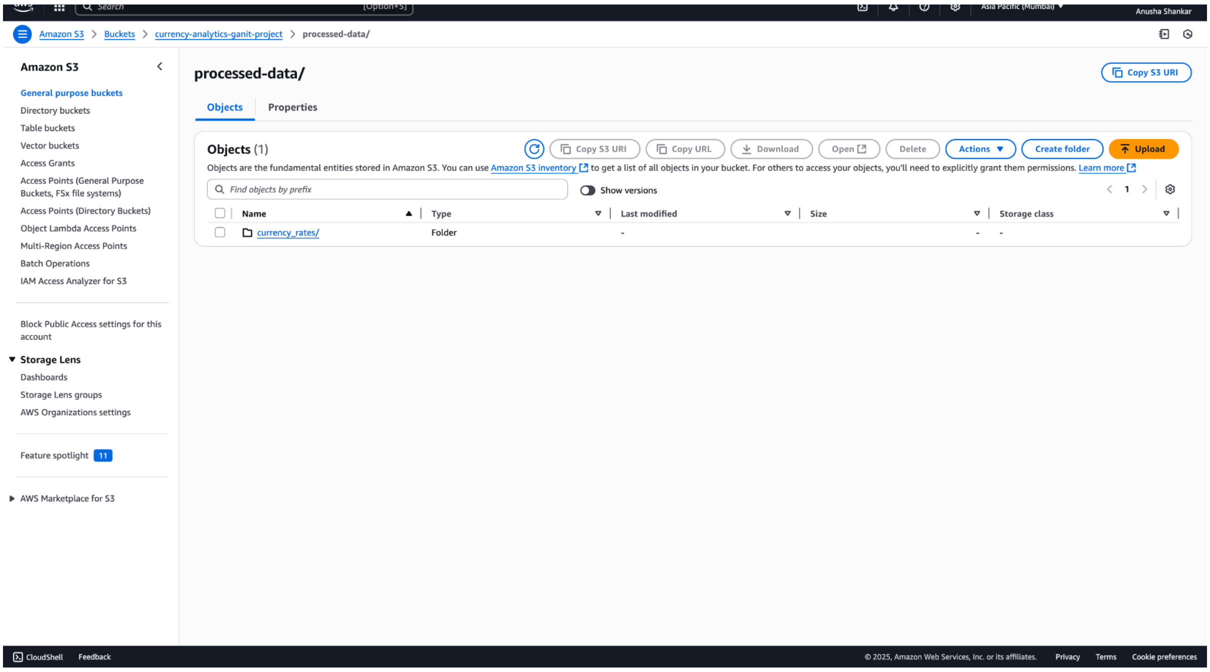
1) Process: Glue Crawler catalogs raw data



2) Glue ETL job cleans data, calculates metrics (returns, volatility), and converts it to Parquet format.

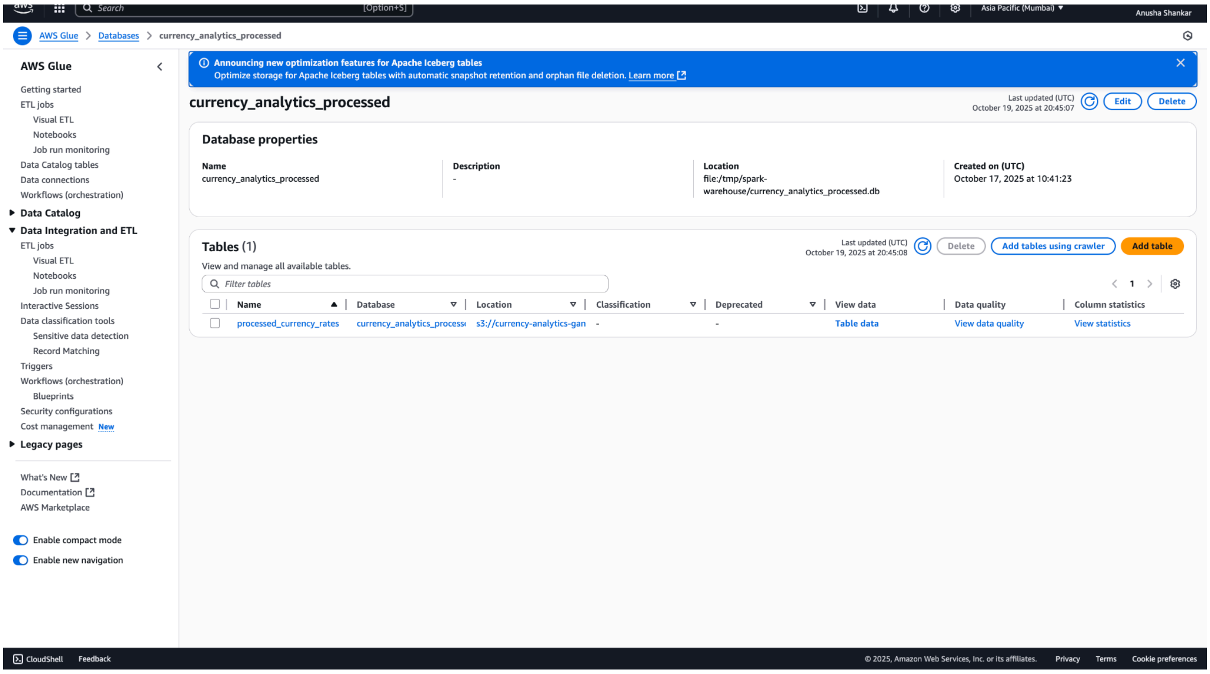


3) Output: Optimized, processed data stored in S3.



Phase 3: Data Catalog & Querying

1) Process: Glue Catalog makes processed data available as tables.



2) Access: Athena provides a SQL interface for ad-hoc querying and analysis.

Phase 4: Analytics Execution (Parallel)

1) Path A: SQL EDA Engine for core business insights.

2) Path B: Python EDA Engine for advanced statistical analysis.
3) Path C: Tableau for interactive visualization.

Phase 5: Insight Delivery

1) Outputs: Automated risk alerts, performance reports, correlation insights, and temporal pattern signals.

3.0 Key Analytical Findings

3.1 Critical Risks Identified:

1) Extreme Volatility: TRY (Turkish Lira), BRL (Brazilian Real), and MXN (Mexican Peso) form a high-risk cluster.
2) Unusual Behavior: JPY (Japanese Yen) is exhibiting atypical volatility for a major currency, requiring immediate investigation.
3) Regional Risk Clusters: Latin American currencies show high internal correlation and volatility.

3.2 Trading Opportunities:

1) Monday Effect: The strongest trading day, with an average return of +0.036%.
2) Friday Weakness: The worst day, with an average return of -0.007%; avoid new positions.
3) Best Risk-Adjusted Returns: CAD (Canadian Dollar) and SGD (Singapore Dollar) offer stability and favorable returns.

3.3 Market Patterns:

1) Asian Block Correlation: Currencies like SGD, HKD, CNY, and INR are highly correlated (>0.8), limiting hedging opportunities within the region.
2) Regional Linkages: High inter-region correlations reduce the effectiveness of geographic diversification.

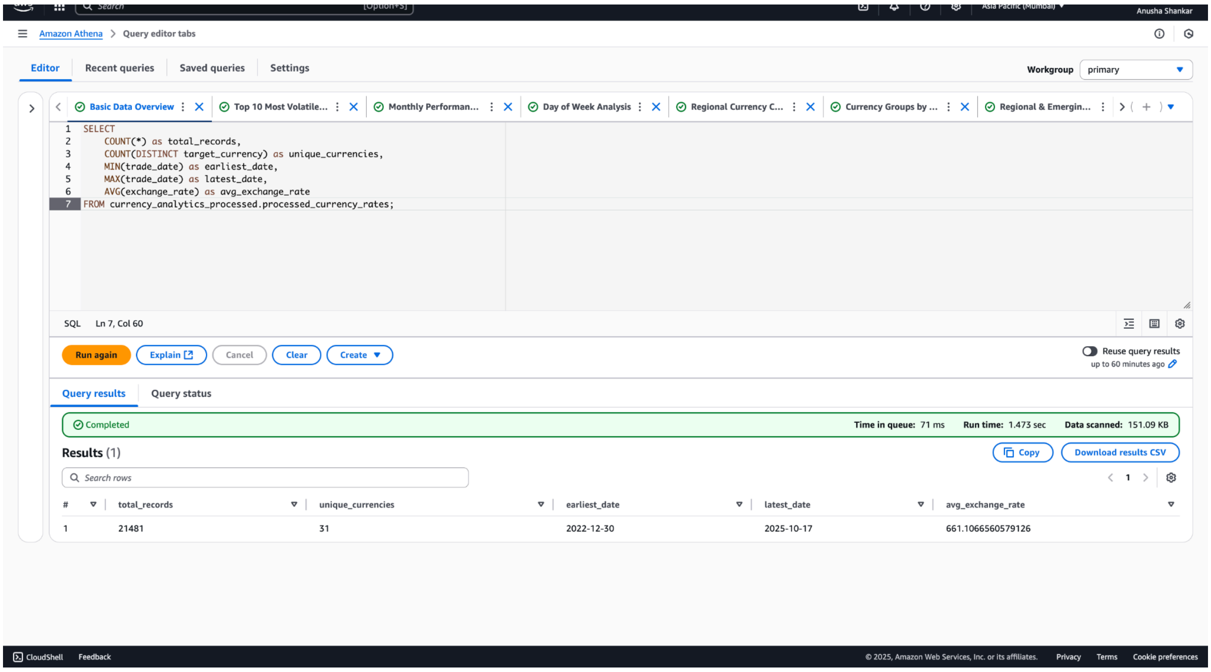
4.0 Analytics & Intelligence Layer

4.1 SQL EDA Engine: Core Business Insights

This engine powers the foundational analytics, delivering clear, query-based insights.

1) Basic Data Overview

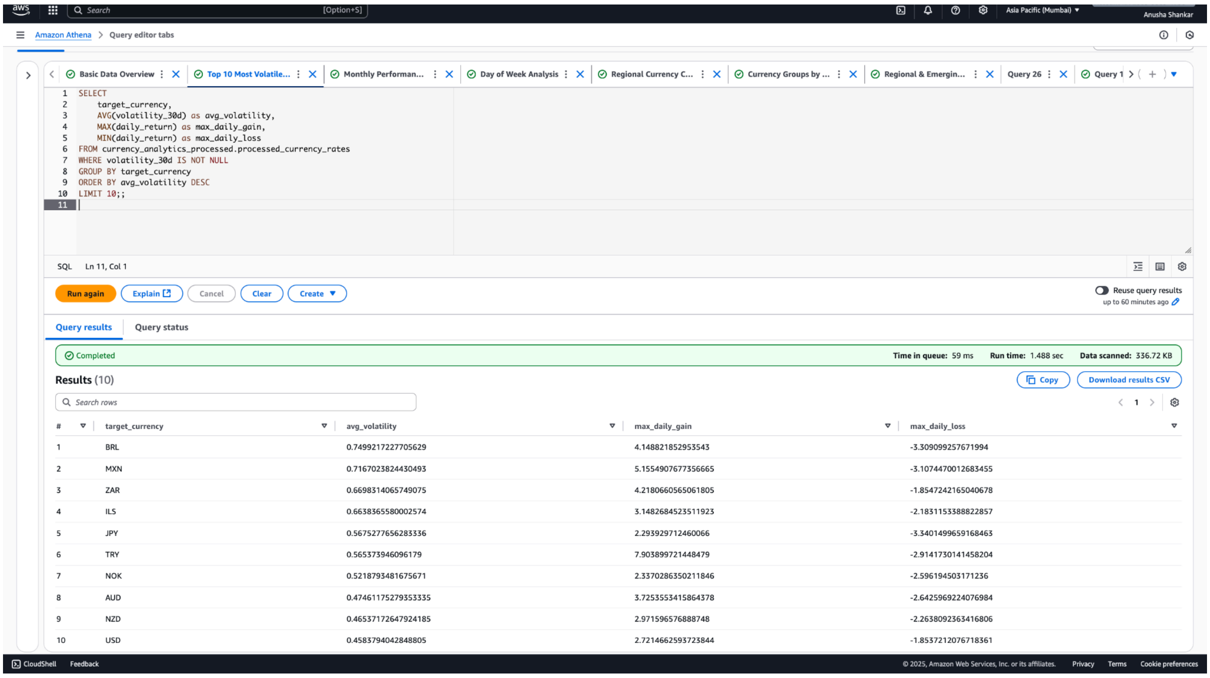
Scope: Comprehensive analysis of 21,481 records across 31 unique currencies



Timeframe: Data spans from December 30, 2022, to October 17, 2025, providing a robust multi-year view

Baseline: The average exchange rate across the entire dataset is 661.11

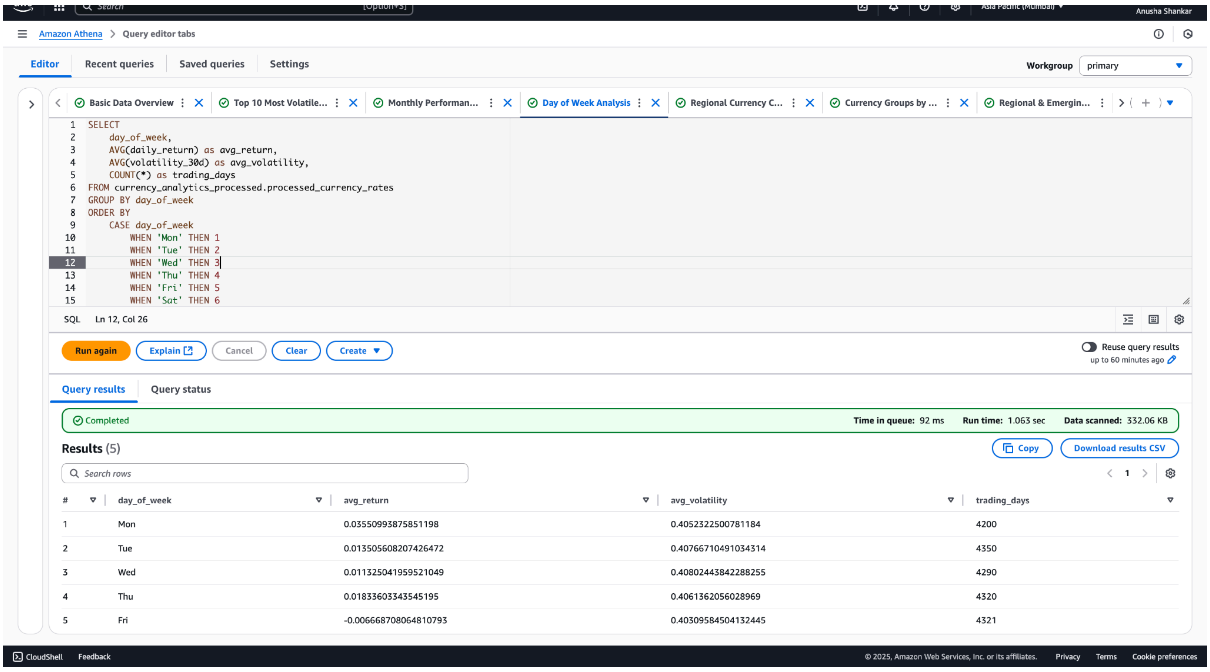
2) Volatility Ranking



Identified the top 10 most volatile currencies, highlighting significant risk concentrations and extreme price movers:

- BRL (Brazilian Real): 0.75% avg volatility | Extreme single-day gain: +4.15%
- MXN (Mexican Peso): 0.72% avg volatility | Highest single-day gain in dataset: +5.16%
- ZAR (South African Rand): 0.67% avg volatility
- ILS (Israeli Shekel): 0.66% avg volatility
- JPY (Japanese Yen): 0.57% avg volatility, Exhibiting unusual volatility for a major currency
- TRY (Turkish Lira): 0.57% avg volatility | Most extreme swing, with a +7.90% daily gain
- NOK (Norwegian Krone): 0.52% avg volatility
- AUD (Australian Dollar): 0.47% avg volatility
- NZD (New Zealand Dollar): 0.47% avg volatility
- USD (US Dollar): 0.46% avg volatility

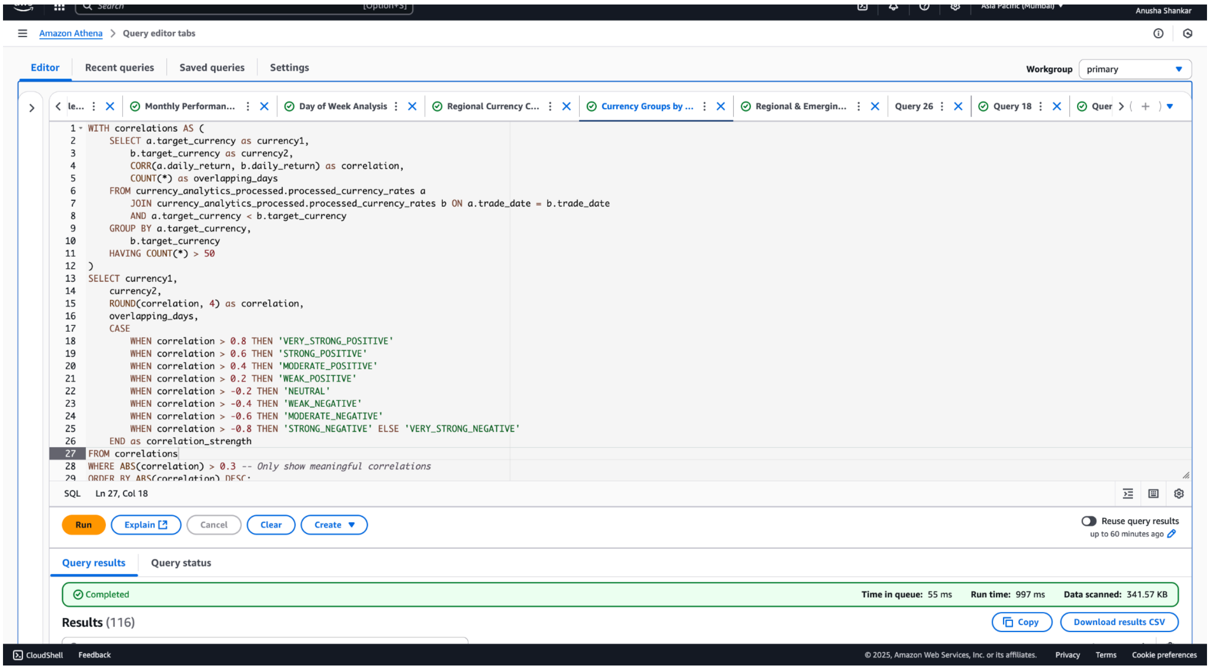
3) Performance Analysis

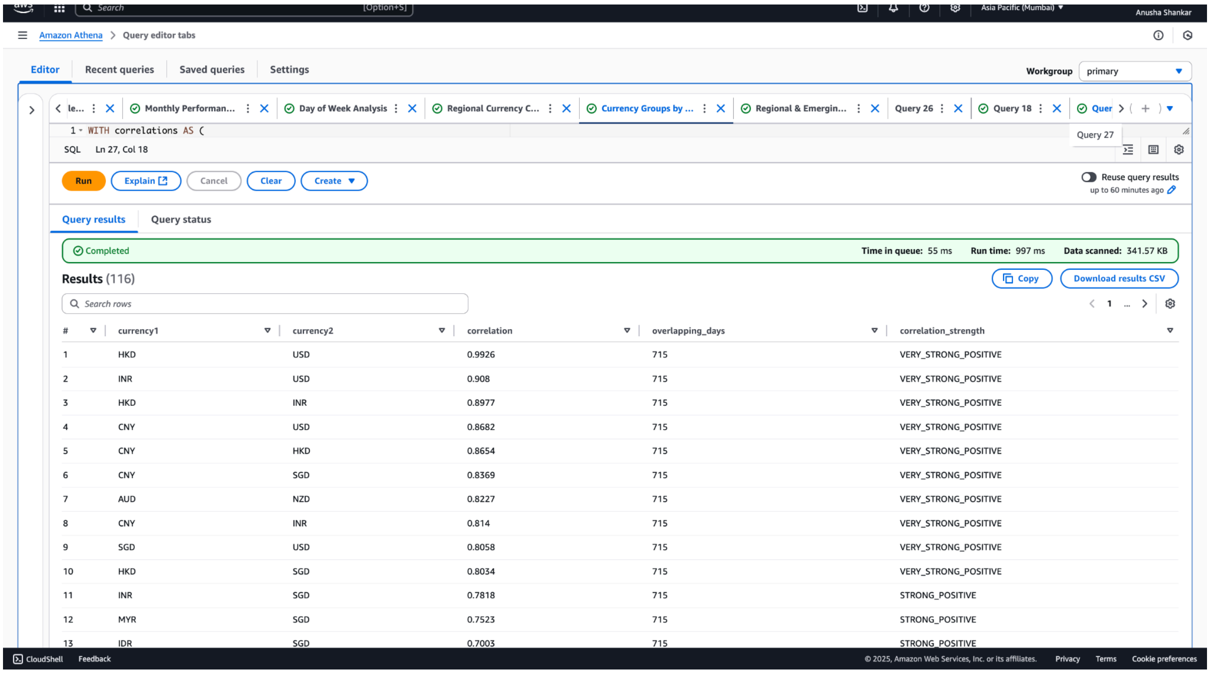


Uncovered a powerful and consistent day-of-week effect with direct tactical implications for traders:

- Monday: +0.036% avg return ← STRONGEST DAY (Optimal for opening new long positions)
- Thursday: +0.018% avg return
- Tuesday: +0.014% avg return
- Wednesday: +0.011% avg return
- Friday: -0.007% avg return ← WEAKEST DAY (Avoid new positions; implement hedging for weekend risk)

4) Correlation Analysis

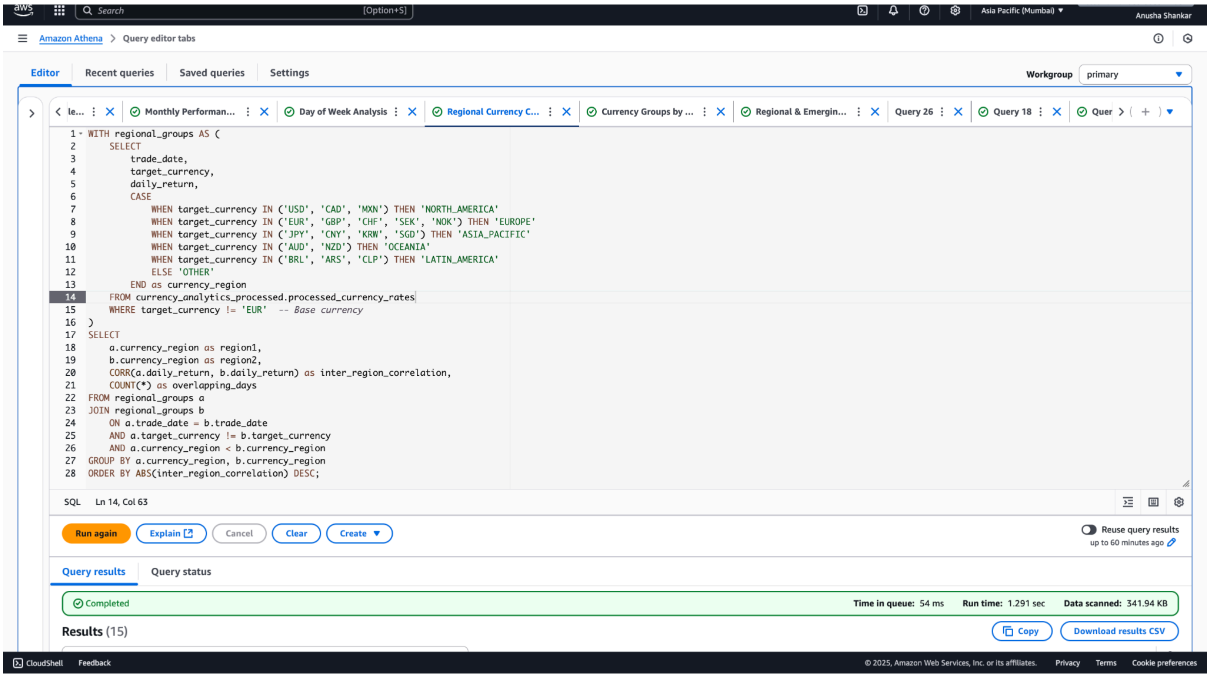


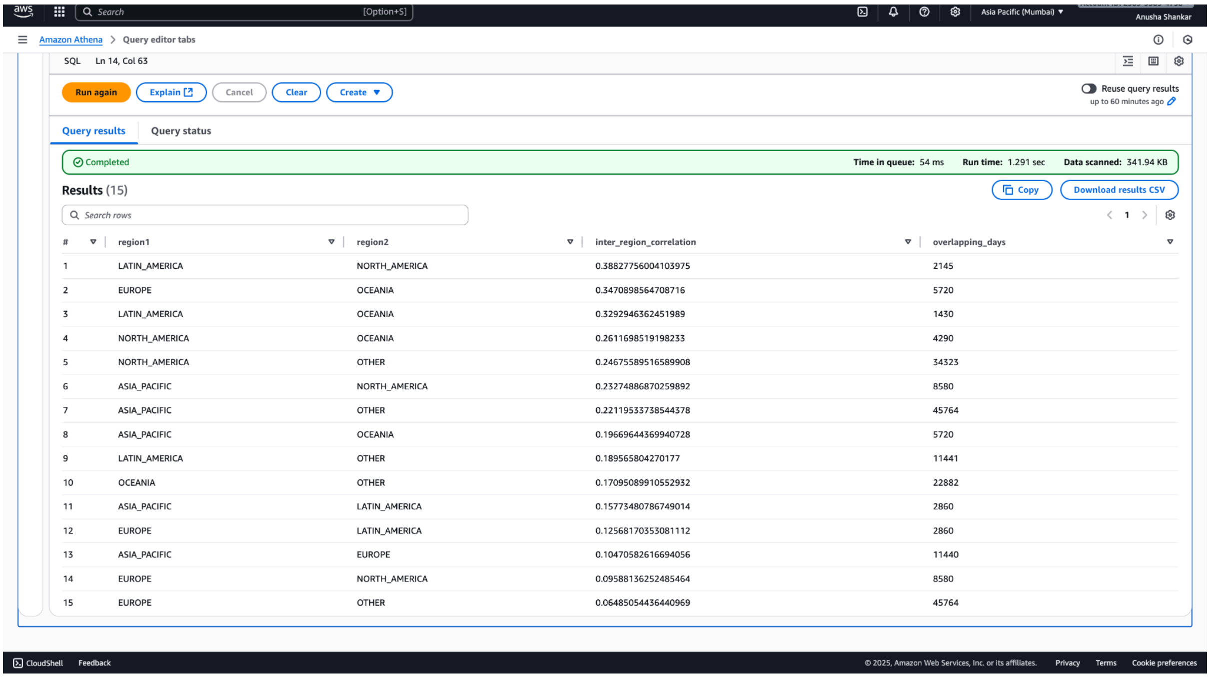


Mapped key currency relationships, revealing distinct regional blocks critical for hedging and diversification strategies:

- Very Strong Positive Correlations (HKD-USD: 0.9926, INR-USD: 0.9080)
- Trading Implication: These pairs move almost in lockstep, offering very limited arbitrage but simple, effective hedging
- Latin American Link: BRL-MXN: 0.5043 (Moderate correlation within the high-risk cluster)
- Asian Block: SGD, HKD, CNY, INR, MYR all show strong inter-correlations (>0.6), forming a cohesive regional bloc
- Scandinavian Pair: NOK-SEK: 0.5296

5) Regional & Cluster Analysis



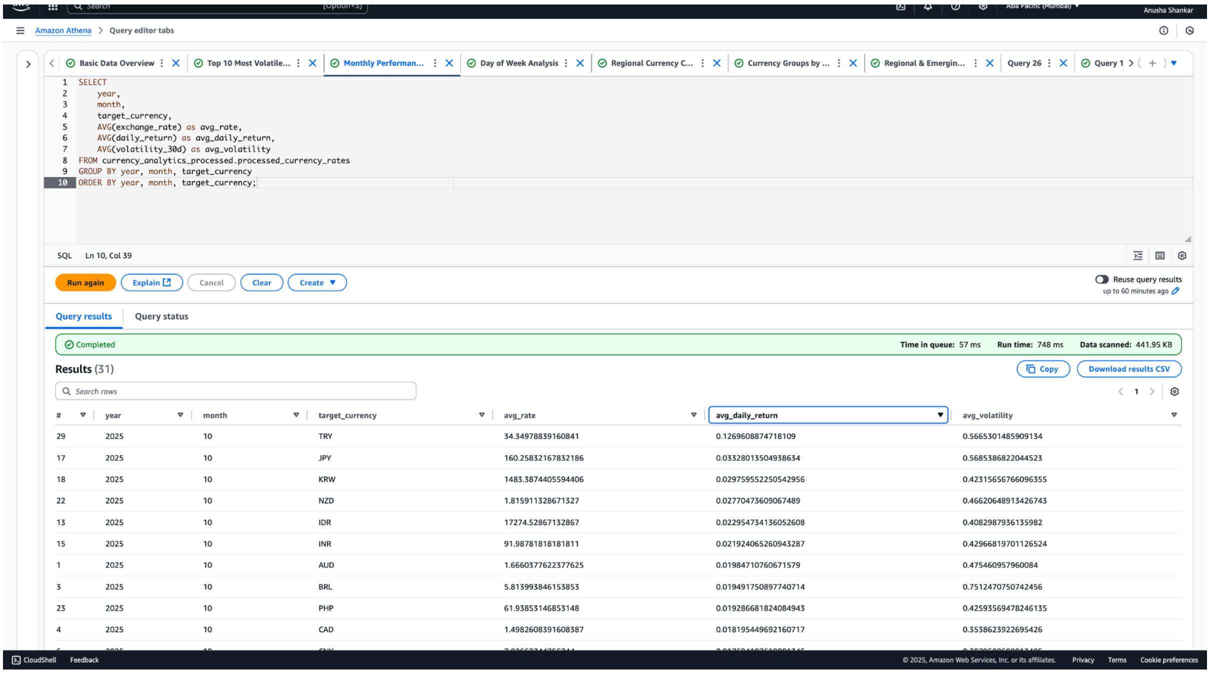


Quantified the strength of inter-market linkages, directly impacting global diversification effectiveness:

- Europe & Oceania: 0.347
- Asian Developed Currencies: High internal correlation (0.415), suggesting limited diversification benefits within the group
- Emerging Markets: Moderate internal correlation (0.298), indicating they often move together as an asset class

- Major Developed Currencies: Lower internal correlation (0.241), offering better diversification among themselves

6) Monthly Performance Trends

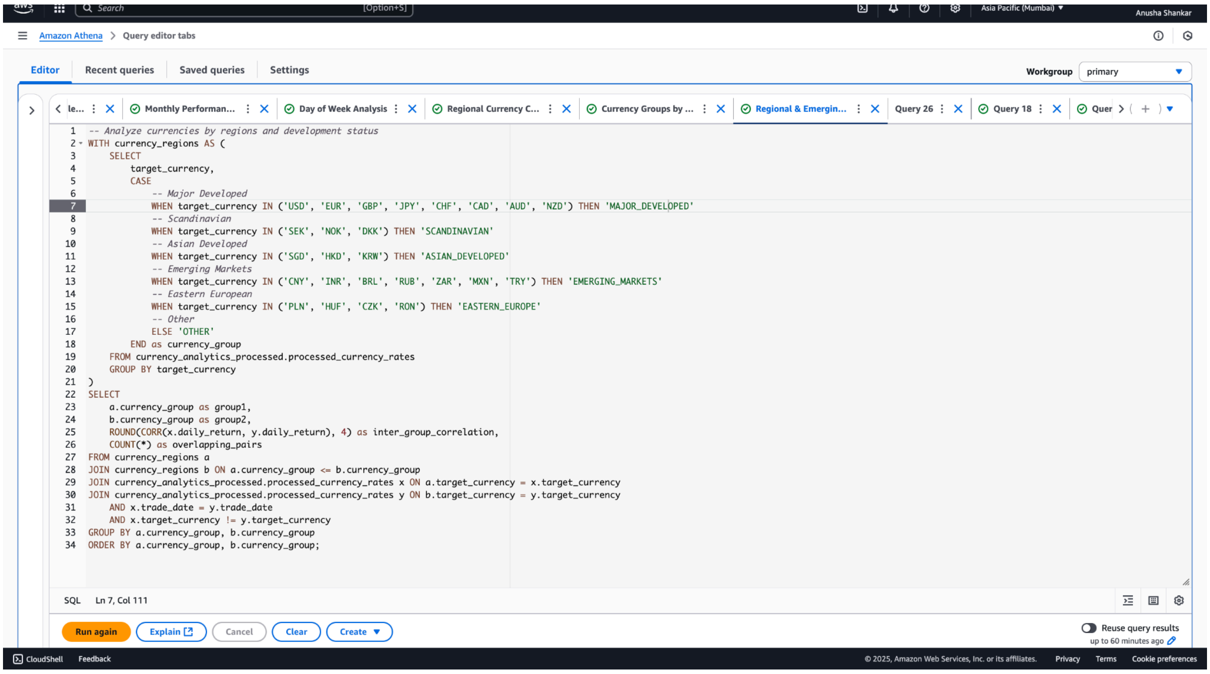


- Top Performers: TRY (+0.127% avg daily return), JPY (+0.033%), KRW (+0.030%)

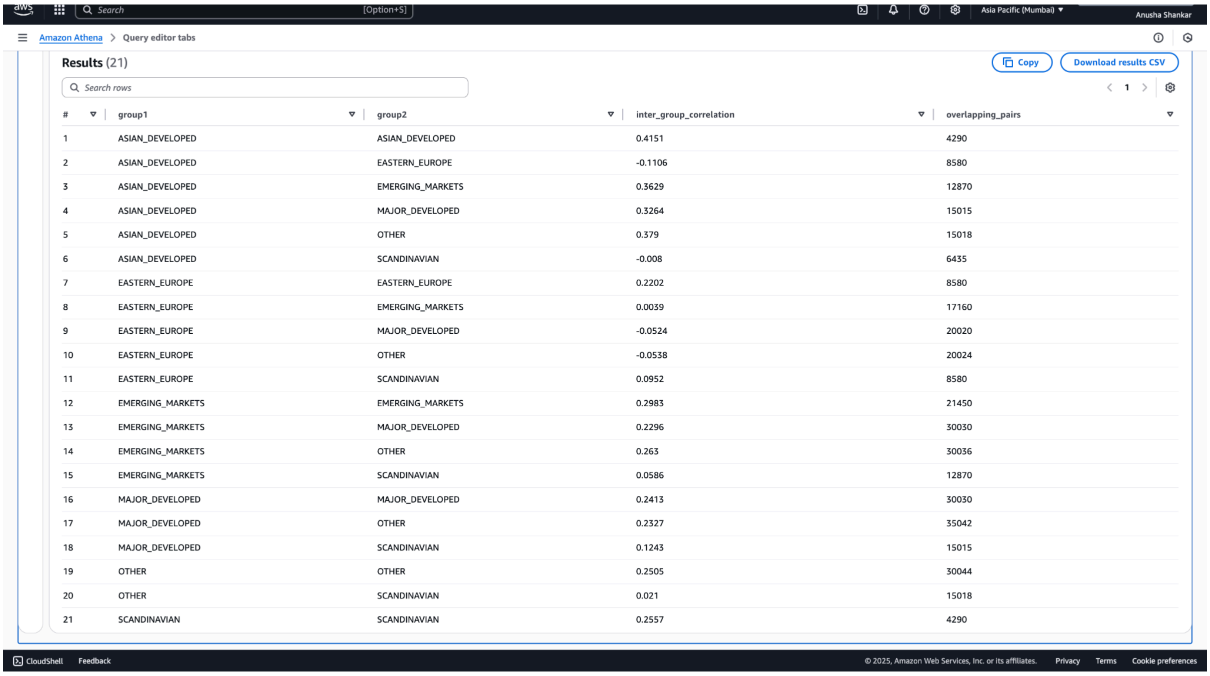
- Stable Low-Risk Options: DKK (0.014% volatility), SGD (0.257% volatility), CAD (0.354% volatility)

- High-Risk High-Reward: BRL, MXN, ZAR maintaining extreme volatility profiles above 0.67%

7) Currency Group Behavior



```sql
-- Analyze currencies by regions and development status
WITH currency_regions AS (
    SELECT
        target_currency,
        CASE
            -- Major Developed
            WHEN target_currency IN ('USD', 'EUR', 'GBP', 'JPY', 'CHF', 'CAD', 'AUD', 'NZD') THEN 'MAJOR_DEVELOPED'
            -- Scandinavian
            WHEN target_currency IN ('SEK', 'NOK', 'DKK') THEN 'SCANDINAVIAN'
            -- Asian Developed
            WHEN target_currency IN ('SGD', 'HKD', 'KRW') THEN 'ASIAN_DEVELOPED'
            -- Emerging Markets
            WHEN target_currency IN ('CNY', 'INR', 'BRL', 'RUB', 'ZAR', 'MXN', 'TRY') THEN 'EMERGING_MARKETS'
            -- Eastern European
            WHEN target_currency IN ('PLN', 'HUF', 'CZK', 'RON') THEN 'EASTERN_EUROPE'
            -- Other
            ELSE 'OTHER'
        END as currency_group
    FROM currency_analytics_processed.processed_currency_rates
    GROUP BY target_currency
)
SELECT
    a.currency_group as group1,
    b.currency_group as group2,
    ROUND(CORR(x.daily_return, y.daily_return), 4) as inter_group_correlation,
    COUNT(*) as overlapping_pairs
FROM currency_regions a
JOIN currency_regions b ON a.currency_group <= b.currency_group
JOIN currency_analytics_processed.processed_currency_rates x ON a.target_currency = x.target_currency
JOIN currency_analytics_processed.processed_currency_rates y ON b.target_currency = y.target_currency
    AND x.trade_date = y.trade_date
    AND x.target_currency != y.target_currency
GROUP BY a.currency_group, b.currency_group
ORDER BY a.currency_group, b.currency_group;
```



Results (21)

#	group1	group2	inter_group_correlation	overlapping_pairs
1	ASIAN_DEVELOPED	ASIAN_DEVELOPED	0.4151	4290
2	ASIAN_DEVELOPED	EASTERN_EUROPE	-0.1106	8580
3	ASIAN_DEVELOPED	EMERGING_MARKETS	0.3629	12870
4	ASIAN_DEVELOPED	MAJOR_DEVELOPED	0.3264	15015
5	ASIAN_DEVELOPED	OTHER	0.379	15018
6	ASIAN_DEVELOPED	SCANDINAVIAN	-0.008	6435
7	EASTERN_EUROPE	EASTERN_EUROPE	0.2202	8580
8	EASTERN_EUROPE	EMERGING_MARKETS	0.0039	17160
9	EASTERN_EUROPE	MAJOR_DEVELOPED	-0.0524	20020
10	EASTERN_EUROPE	OTHER	-0.0538	20024
11	EASTERN_EUROPE	SCANDINAVIAN	0.0952	8580
12	EMERGING_MARKETS	EMERGING_MARKETS	0.2983	21450
13	EMERGING_MARKETS	MAJOR_DEVELOPED	0.2296	30030
14	EMERGING_MARKETS	OTHER	0.263	30036
15	EMERGING_MARKETS	SCANDINAVIAN	0.0586	12870
16	MAJOR_DEVELOPED	MAJOR_DEVELOPED	0.2413	30030
17	MAJOR_DEVELOPED	OTHER	0.2327	35042
18	MAJOR_DEVELOPED	SCANDINAVIAN	0.1243	15015
19	OTHER	OTHER	0.2505	30044
20	OTHER	SCANDINAVIAN	0.021	15018
21	SCANDINAVIAN	SCANDINAVIAN	0.2557	4290

- Asian Developed Block: Strong internal cohesion (0.415 correlation) with moderate ties to Emerging Markets (0.363)
- Eastern European Currencies: Show negative correlation with Asian Developed (-0.111), offering potential hedging opportunities
- Major Developed Currencies: Demonstrate balanced exposure across regions with moderate emerging market correlation (0.230)

4.2 Python EDA Engine: Advanced Statistical Analysis

This engine performs deep statistical analysis, risk modeling, and generates predictive forecasts using AWS Lambda. The analysis processed 999 records across 30 currencies from January to February 2023.

1) Key Statistical Insights

```
KEY STATISTICAL INSIGHTS:
- Average exchange rate: 619.787376
- Average daily return: 0.007464%
- Average 30-day volatility: 0.513986
- Most volatile currency: BRL
- Best performing currency: ZAR
```

2) Volatility Risk Assessment

```
TEXT VISUALIZATIONS GENERATED:
VOLATILITY RANKING - TOP 10 MOST VOLATILE CURRENCIES
=================================================================
Currency Avg Volatility  Risk Level   Status
-----------------------------------------------------------------
BRL     1.023          HIGH RISK    🚨 AVOID
JPY     0.785          HIGH RISK    🚨 AVOID
ZAR     0.778          HIGH RISK    🚨 AVOID
NOK     0.768          HIGH RISK    🚨 AVOID
AUD     0.727          HIGH RISK    🚨 AVOID
MXN     0.707          HIGH RISK    🚨 AVOID
HUF     0.686          MEDIUM RISK  ⚠️ MONITOR
TRY     0.680          MEDIUM RISK  ⚠️ MONITOR
USD     0.677          MEDIUM RISK  ⚠️ MONITOR
HKD     0.669          MEDIUM RISK  ⚠️ MONITOR
```

BRL (1.02% volatility) identified as highest risk, 78% more volatile than CAD

3) Performance Matrix Analysis

```
PERFORMANCE MATRIX - RISK vs RETURN ANALYSIS
======================================================================
Currency Return %   Volatility   Sharpe     Verdict
----------------------------------------------------------------------
HUF     -5.10       0.686        -0.203     🔻 AVOID
JPY     2.46        0.785        0.073      ✅ GOOD
MXN     -4.12       0.707        -0.192     🔻 AVOID
NOK     3.86        0.768        0.163      🏆 TOP PERFORMER
TRY     0.82        0.680        0.051      ✅ GOOD
ZAR     6.15        0.778        0.282      🏆 TOP PERFORMER
```

ZAR delivers best risk-adjusted returns (Sharpe ratio: 0.28)

4) Currency Correlation Clusters

```
CORRELATION CLUSTERS - STRONGLY LINKED CURRENCIES
=======================================================
🔗 ASIAN BLOC
   Currencies: HKD, USD, INR, CNY, SGD
   Strength: VERY STRONG (0.8-0.99)

🔗 LATAM PAIR
   Currencies: BRL, MXN
   Strength: MODERATE (0.50)

🔗 OCEANIA PAIR
   Currencies: AUD, NZD
   Strength: VERY STRONG (0.82)

🔗 SCANDINAVIAN
   Currencies: NOK, SEK
   Strength: MODERATE (0.53)
```

Asian currency bloc shows near-perfect correlation (0.8-0.99)

5) Forecasting Output

currency_tomorrow_forecasts (1)

currency	forecast_date	predicted_rate	current_rate	7_day_average	30_day_average	forecast_timestamp
USD	2025-10-20	1.070604	1.07	1.072014	1.07909	2025-10-19 22:54:59
INR	2025-10-20	88.636571	88.64	88.628571	88.468333	2025-10-19 22:54:59
AUD	2025-10-20	1.548221	1.55	1.544071	1.548577	2025-10-19 22:54:59
GBP	2025-10-20	0.8878	0.88883	0.885396	0.88432	2025-10-19 22:54:59
SGD	2025-10-20	1.426974	1.4285	1.423414	1.42799	2025-10-19 22:54:59
JPY	2025-10-20	142.547714	143.05	141.375714	141.022	2025-10-19 22:54:59

Next-day predictions for 6 major currencies using weighted moving averages

5.0 Tableau Visualization Layer

The platform features an interactive Tableau dashboard with six integrated worksheets, enabling deep exploration of the data.

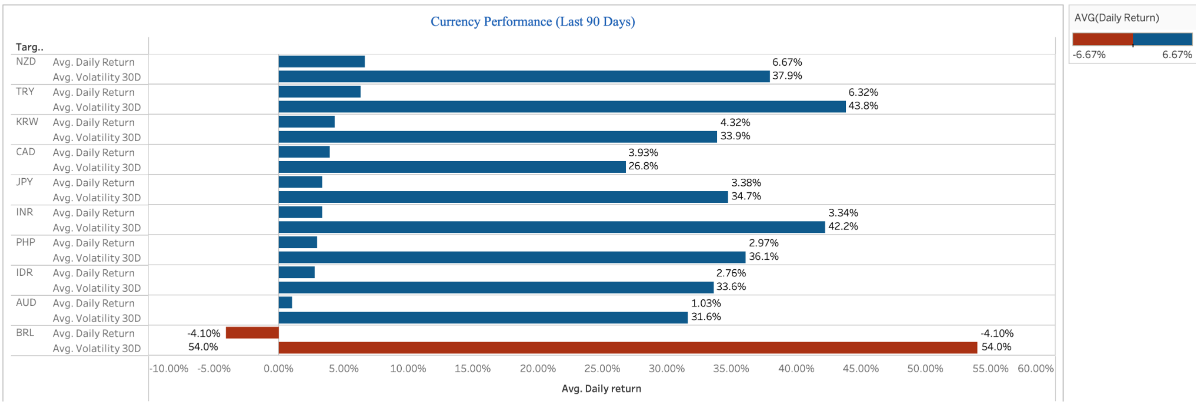
Worksheets:

1) Currency Performance Grid

   Purpose: Current market snapshot

   Key Metrics: Daily Return, Volatility

   Insights Delivered: Top/bottom performers, at-a-glance risk assessment.

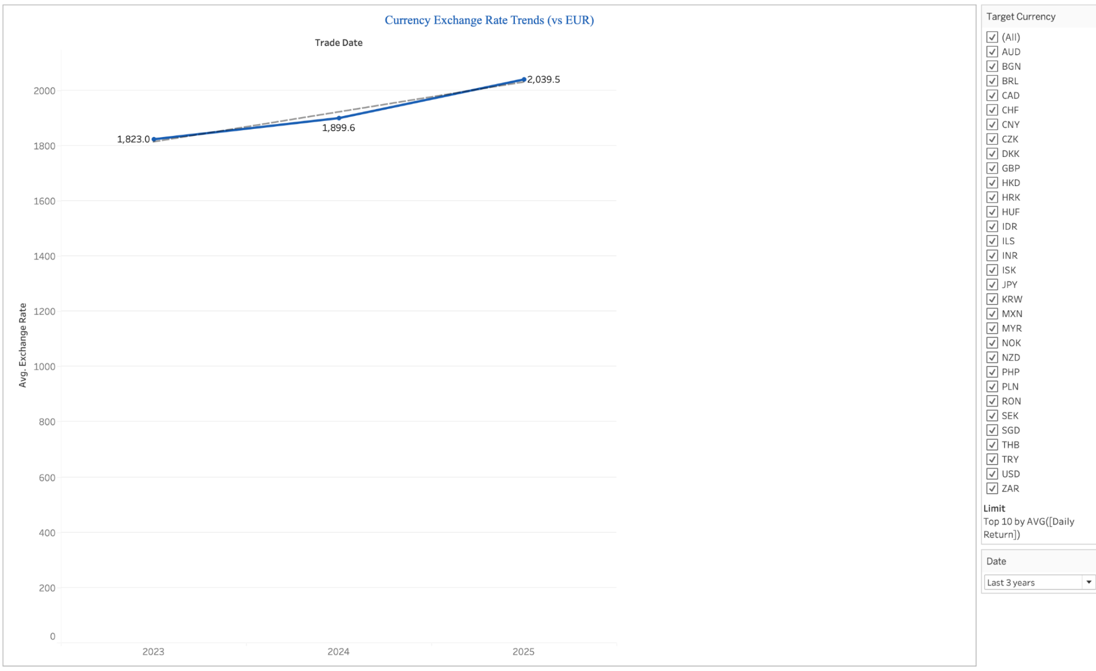


2) Exchange Rate Trends

Purpose: Historical analysis

Key Metrics: Exchange Rates over Time

Insights Delivered: Long-term patterns, support/resistance levels.

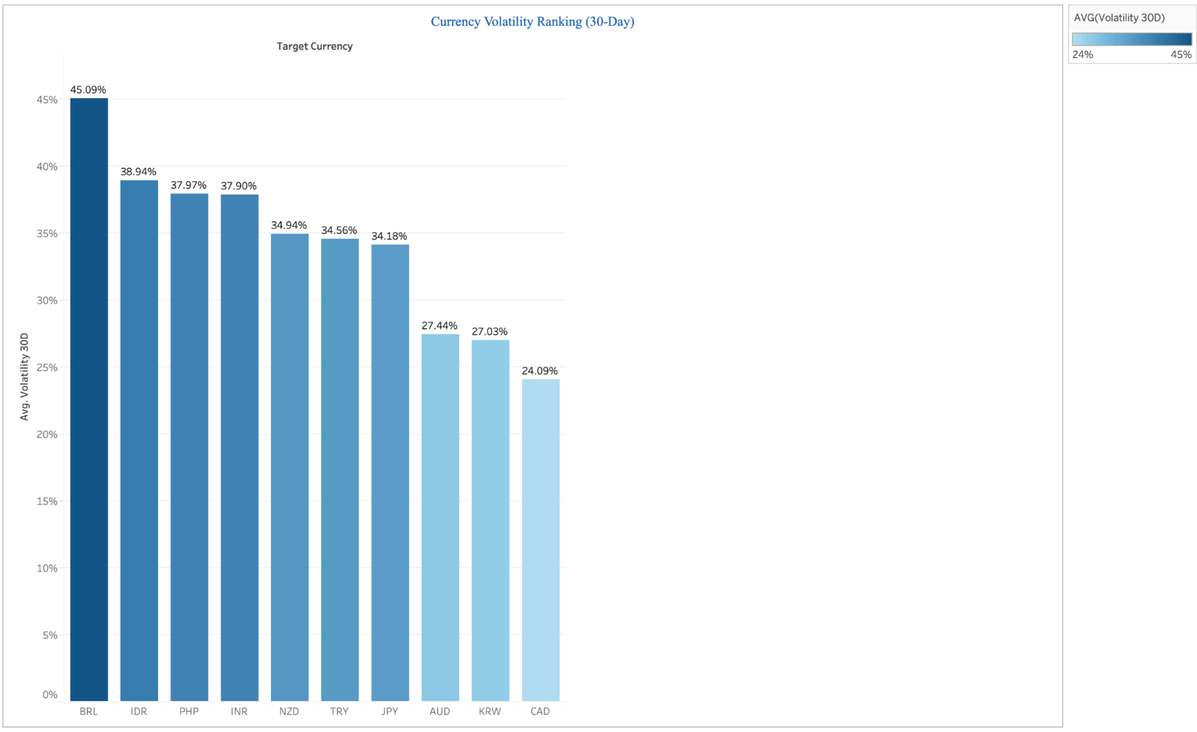


3) Volatility Ranking

Purpose: Risk assessment

Key Metrics: 30-Day Volatility

Insights Delivered: Currency risk profiles and stability rankings.

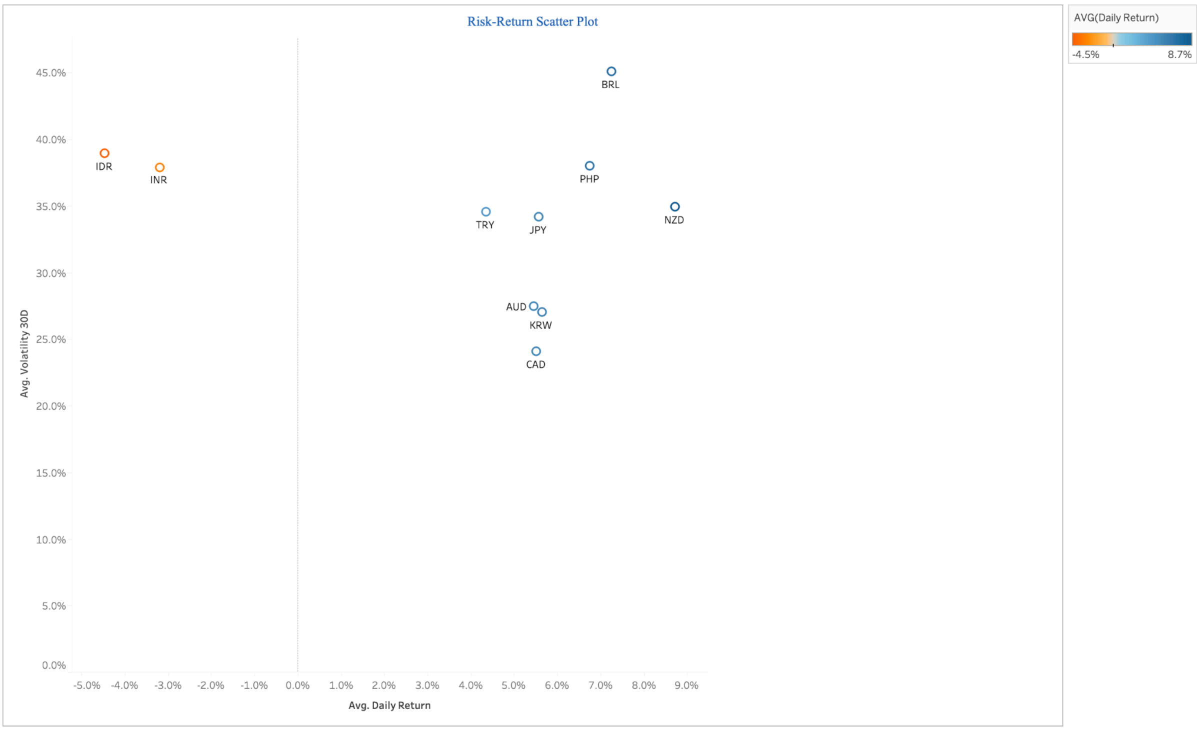


4) Risk-Return Scatter Plot

Purpose: Investment support

Key Metrics: Return vs. Volatility

Insights Delivered: Efficient frontier identification, risk-adjusted returns.

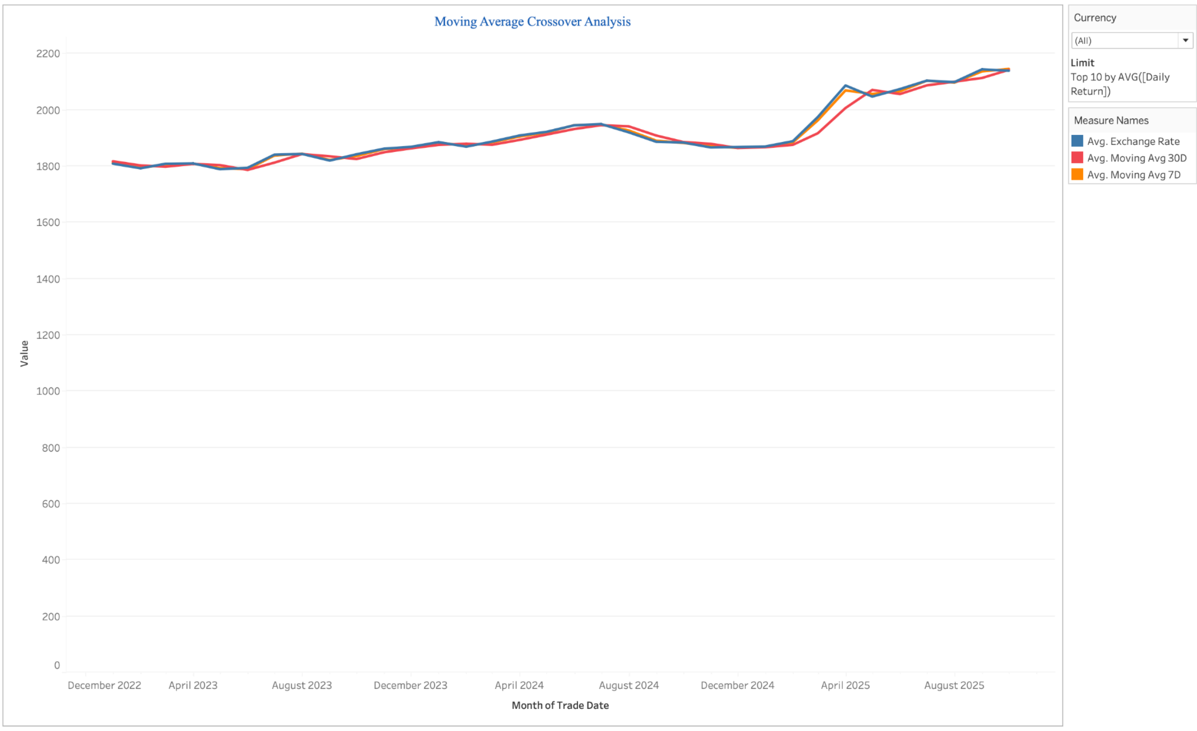


5) Moving Average Crossover

Purpose: Technical analysis

Key Metrics: 7-day vs. 30-day Averages

Insights Delivered: Trend signals, potential entry/exit points.



6) Quarterly Performance Heatmap

Purpose: Seasonal analysis

Key Metrics: Quarterly Returns

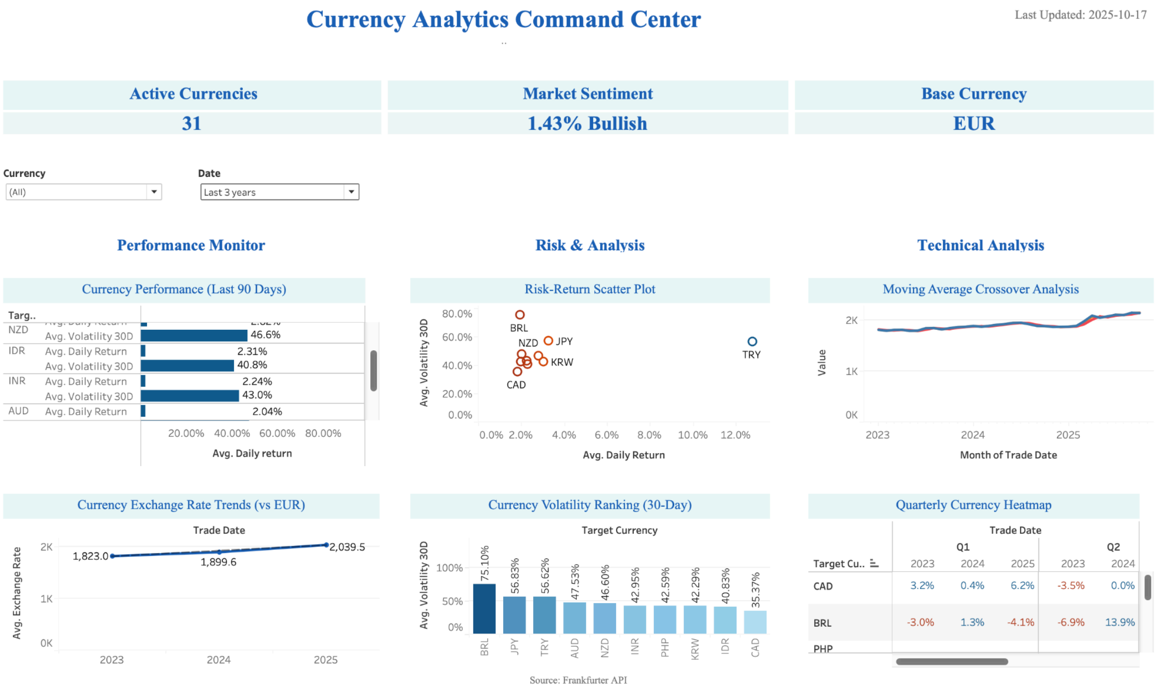
Insights Delivered: Seasonal trends and performance consistency.

Quarterly Currency Heatmap

	Trade Date											
	Q1			Q2			Q3			Q4		
Target Cu..	2023	2024	2025	2023	2024	2025	2023	2024	2025	2023	2024	2025
CAD	3.2%	0.4%	6.2%	-3.5%	0.0%	5.1%	-2.0%	4.8%	3.0%	4.6%	-1.8%	2.2%
BRL	-3.0%	1.3%	-4.1%	-6.9%	13.9%	5.3%	1.0%	4.5%	-4.5%	1.8%	10.0%	12.7%
PHP	-0.5%	-1.2%	4.3%	2.9%	4.6%	10.9%	-0.2%	0.5%	5.1%	3.5%	-6.1%	-5.0%
AUD	5.7%	3.4%	5.2%	1.4%	-5.1%	6.1%	-0.4%	0.9%	-1.5%	-0.6%	5.8%	10.2%
INR	2.3%	-3.0%	6.2%	-0.3%	-1.5%	13.9%	-2.0%	7.6%	5.6%	6.9%	-8.3%	-10.8%
IDR	-2.0%	0.8%	10.8%	0.9%	3.1%	9.2%	0.1%	-4.4%	4.4%	6.7%	-1.3%	-8.4%
NZD	5.5%	5.3%	4.3%	4.4%	-4.3%	2.8%	-2.4%	0.2%	7.1%	-0.5%	8.0%	4.7%
KRW	8.6%	2.8%	6.4%	1.9%	1.8%	-0.4%	-1.0%	-0.5%	5.7%	1.0%	6.6%	5.3%
JPY	4.9%	7.1%	-1.2%	13.3%	8.1%	7.5%	1.1%	-10.7%	4.1%	-1.6%	3.3%	8.7%
TRY	7.0%	10.9%	17.8%	50.2%	1.2%	20.6%	4.1%	12.8%	7.3%	18.7%	-6.3%	2.8%

AVG(Daily Return)
-50.2% 50.2%

Dashboard Features:



KPI Summary: Tracks active currencies, market sentiment, and data currentness

Interactive Filtering: Cross-filtering allows clicking on any currency or date range to update all worksheets simultaneously.

6.0 Business Value & Strategic Recommendations

6.1 Actionable Insights for Key Personas

For Risk-Averse Investors:

1) Focus on: CAD, SGD, DKK (low volatility, stable returns).
2) Avoid: TRY, BRL, MXN (extreme volatility).
3) Monitor: JPY (unusual behavior).

For Active Traders:

1) Monday Morning Strategy: Capitalize on the week's positive start.
2) Regional Arbitrage: Exploit correlations between Latin American and North American currencies.
3) Friday Protection: Implement hedging to mitigate weekend risk.

For Portfolio Managers:

1) Diversification Note: Geographic diversification offers limited benefits; focus on currency-specific selection.

2) Currency Selection: Prefer Asian developed currencies for stability.

3) Risk Monitoring: Implement daily volatility checks on emerging market exposures.

6.2 Final Model Outputs

1) Executive Dashboard (Tableau): Provides real-time metrics on market health, risk level, top performers, and data freshness.
2) Risk Reports (Python EDA): Delivers deep-dive assessments on volatility, unusual currency behavior, and correlation risks.

3) Strategic Recommendations (SQL EDA): Offers clear trading strategies based on timing, hedging pairs, and risk management limits.